

Sorting Four-Sided Shapes by What They Share

Sorting four-sided shapes by square corners, by side lengths, and by parallel sides, and reading a shape-sort graph. Ms. Diaz's class made a set of shape cards. Every card has four straight sides.

- A **square corner** is a corner shaped like the corner of a piece of paper. Use the corner of your paper to test one.
- Two sides are **parallel** when they run the same way and would never bump into each other, like two rails of a train track.
- Shapes are drawn the true size they were cut, so you can compare side lengths by looking and by measuring.

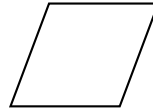
1. Circle every card below that has **four square corners**. Then write how many cards you circled.



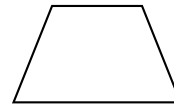
A



B



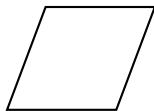
C



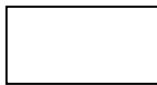
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Answer: _____

2. Circle every card below whose **four sides are all the same length**. Then write how many cards you circled.



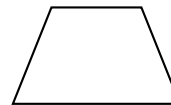
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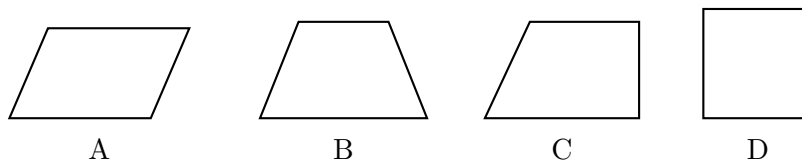
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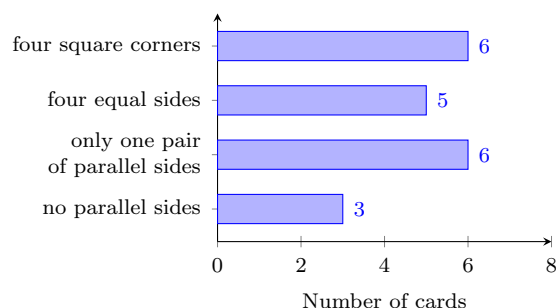
Answer: _____

3. A card goes in Ms. Diaz’s “one pair” box when it has **exactly one pair of parallel sides**. Circle every card that belongs in the “one pair” box, then write how many cards you circled.



Answer: _____

4. The class sorted their whole set of cards into four groups and made this graph. **(a)** How many cards did the class sort in all? **(b)** How many more cards went in the “four square corners” group than in the “no parallel sides” group?



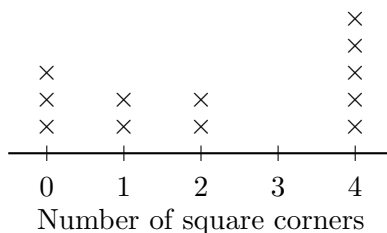
Answer: _____

- 5.** Rosa says the card below belongs in the rectangle group. Ben says the same card belongs in the square group. Their teacher says they are both right. Use the words *sides* and *corners* to explain how one card can belong in both groups.



Answer: _____

- 6.** Each $\times$ on the line plot stands for one shape card. The number below tells how many square corners that card has.



- (a)** How many shape cards are on the line plot in all? **(b)** Which number of square corners happened most often? **(c)** No card sits above the mark for three. Explain why a shape with four straight sides can never have exactly three square corners.

Answer: _____

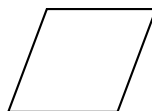
7. Nia made a group and named it “all four sides the same length.” Circle every card below that belongs in Nia’s group, then write how many cards belong in it.



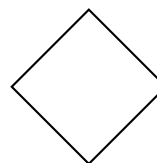
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Answer: _____

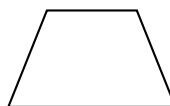
8. Kim is filling the “two pairs of parallel sides” box. She looks at the four cards below and says that all 4 of them belong in it. Kim is wrong. Circle the cards that really belong, write how many there are, and write the letter of the card Kim should take back out.



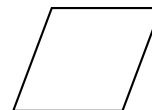
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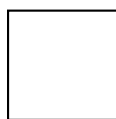
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Answer: _____

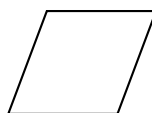
9. Some cards belong in *two* groups at once. Circle every card below that has **four square corners** *and* **four sides the same length**, then write how many cards you circled.



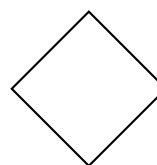
A



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Answer: _____

10. On the dot paper below, draw one four-sided shape with *exactly one* pair of parallel sides, and beside it draw one four-sided shape with *two* pairs of parallel sides. Then write one sentence naming something both of your shapes share.

